

Predicting Cheque Dishonor Probabilities

An Automated Machine Learning Solution for Risk Mitigation

The Operational Challenge

Identifying financial risk in a high-volume, manual transaction environment.

Legacy Tracking Limitations



Reactive Nature

Teams respond only after a bounce occurs, causing significant collection delays and interest loss.



Behavioral Blindspots

Critical trends in Agent performance and Insured history are buried in massive "RAW Data" spreadsheets.



Subjective Risk

Decisions rely on intuition rather than statistical probability, leading to inconsistent risk exposure.

| The ML Framework

Our solution implements a **Random Forest Classifier** to process active transactions through an automated cloud bridge.

By mapping historical resolutions to live data, the application assigns a real-time risk score to every outstanding instrument.



| Integrated Data Ecosystem

Historical "RAW Data"

A repository of thousands of historical cheque statuses used to train the ensemble model on behavioral patterns.

Live "dishonorPredict"

The active deployment tab where probabilities are updated daily via an automated Python script.

The Predictive Methodology

Analyzing feature importance and statistical significance of financial signals.

Model Feature Importance



Random Forest feature impact analysis for Dishonor Prediction.

Defining "Days in Limbo"



Real-Time Risk Escalation

This dynamic metric tracks the aging of a cheque from collection to present day. It acts as a primary alarm: as time passes without clearing, the risk probability escalates non-linearly.

- Day 1-2: Standard clearing cycle.
- Day 3-5: High-risk detection window.
- Day 7+: Critical probability plateau.

Cloud System Architecture

Secure Python-Sheets Bridge

The system utilizes a **Google Service Account** (JSON Key) to bypass browser-based logins, ensuring 100% reliable background execution.

Hosted on cloud runners, the script executes every morning, scanning the RA 16 spreadsheet and writing back probabilities side-by-side.



Performance Metrics

Capability	Manual Tracking	Predictive AI Application
Risk Identification	After dishonor occurs	24-48 hours pre-resolution
Decision Speed	Manual weekly reviews	Automated daily updates
Data Depth	Current status only	Comprehensive multi-factor analysis
Resource Usage	High administrative hours	Autonomous background runner

Deployment Roadmap



Data Sync

API connection & historical cleanup



ML Training

Feature engineering & model
validation



Automation

Scheduling via Cloud Runner



Integration

Full-scale operational dashboarding

Any Questions?

Proactive Risk Management through Data Science

Cheque Dishonor Predict Application | Version 1.0

Image Sources



<https://media.istockphoto.com/id/2245670697/photo/abstract-visualization-of-ai-processing-big-data.jpg?s=612x612&w=0&k=20&c=hrIQkVbAJSYZkIF0y7Eeo94LCuONa9aXuTcBFVp69Xg=>

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